

Model Lab

Every prior, same slate — 2026-10-01

SERVED PRIOR

Served (full season)

PRIORS COMPARED

11

HELD-OUT PA

36,227

Held-Out Scores

Fitted on plate appearances before 2026-08-23, scored on the 36,227 that came after (actual HR rate 3.01%). No outcome in the test window was visible during fitting.

Metrics are per plate appearance; the probabilities lower down are per game.

Read AUC first — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5856	1.60×	0.02913	0.13411	0.00506
KNN comps Prior is the pooled rate of the nearest comparables.	0.5969	1.52×	0.02910	0.13334	0.00401
SVR Support-vector regression on the same feature matrix.	0.5944	1.48×	0.02915	0.13413	0.00683
Random forest Forest regression; a learned similarity metric.	0.5934	1.51×	0.02914	0.13374	0.00595
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5969	1.52×	0.02910	0.13334	0.00401
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5969	1.52×	0.02910	0.13334	0.00401
Linear regression Ordinary least squares on the same features.	0.5955	1.53×	0.02912	0.13364	0.00430
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5961	1.51×	0.02915	0.13380	0.00637
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5960	1.56×	0.02913	0.13362	0.00586
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5929	1.48×	0.02916	0.13390	0.00699

Blend weights (forest variant): KNN 1.00, SVR 0.00, RF 0.00

Shrinkage K per prior: league 251, knn 251, svr 251, rf 251, core 251, ensemble 251, linear 251, logistic 251, nn 251, xgb 251 — estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-Sample Subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (4,012 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

Prior	AUC	Top-decile lift	Brier	Log loss	Cal. RMSE
Flat league League HR rate as the prior for every hitter.	0.5427	1.26×	0.02316	0.11268	0.01017
KNN comps Prior is the pooled rate of the nearest comparables.	0.5682	1.16×	0.02322	0.11306	0.01507
SVR Support-vector regression on the same feature matrix.	0.5824	1.26×	0.02355	0.11779	0.02191
Random forest Forest regression; a learned similarity metric.	0.5888	1.26×	0.02346	0.11479	0.02007
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5682	1.16×	0.02322	0.11306	0.01507
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5682	1.16×	0.02322	0.11306	0.01507
Linear regression Ordinary least squares on the same features.	0.5746	1.26×	0.02336	0.11521	0.01764
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5805	0.95×	0.02347	0.11513	0.02103
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5793	1.26×	0.02332	0.11393	0.01666
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5916	1.16×	0.02356	0.11562	0.02265

Slate Agreement

Served (full season) · 6/6 shared 2/6 hit

Ronald Acuña Jr.
Matt Olson ✓
Drake Baldwin
Kyle Schwarber
Austin Riley
Michael Harris II ✓

KNN comps · 6/6 shared 2/6 hit

Ronald Acuña Jr.
Matt Olson ✓
Drake Baldwin
Kyle Schwarber
Austin Riley
Michael Harris II ✓

Random forest · 5/6 shared 1/6 hit

Ronald Acuña Jr.
Matt Olson ✓
Drake Baldwin
Kyle Schwarber
Austin Riley
Bryce Harper

Forest blend · 6/6 shared 2/6 hit

Ronald Acuña Jr.
Matt Olson ✓
Drake Baldwin
Kyle Schwarber
Austin Riley
Michael Harris II ✓

Logistic regression · 6/6 shared 2/6 hit

Flat league · 6/6 shared 2/6 hit

21.0%	Ronald Acuña Jr.	20.2%
20.9%	Matt Olson ✓	19.1%
19.2%	Drake Baldwin	18.6%
18.6%	Kyle Schwarber	17.1%
17.1%	Austin Riley	15.8%
15.1%	Michael Harris II ✓	15.2%

SVR · 6/6 shared 2/6 hit

20.9%	Matt Olson ✓	21.1%
20.4%	Ronald Acuña Jr.	21.1%
19.4%	Drake Baldwin	19.3%
18.4%	Kyle Schwarber	19.2%
16.7%	Austin Riley	16.7%
15.1%	Michael Harris II ✓	15.5%

Core (KNN+SVR) · 6/6 shared 2/6 hit

20.9%	Ronald Acuña Jr.	20.9%
20.8%	Matt Olson ✓	20.4%
19.7%	Drake Baldwin	19.4%
18.9%	Kyle Schwarber	18.4%
16.3%	Austin Riley	16.7%
14.9%	Michael Harris II ✓	15.1%

Linear regression · 6/6 shared 2/6 hit

20.9%	Ronald Acuña Jr.	20.8%
20.4%	Matt Olson ✓	20.5%
19.4%	Drake Baldwin	19.4%
18.4%	Kyle Schwarber	18.8%
16.7%	Austin Riley	16.5%
15.1%	Michael Harris II ✓	15.2%

Matt Olson ✓	21.2%	Neural net (torch) · 6/6 shared 2/6 hit	
Ronald Acuña Jr.	20.9%	Ronald Acuña Jr.	21.0%
Kyle Schwarber	19.7%	Matt Olson ✓	20.9%
Drake Baldwin	19.4%	Drake Baldwin	19.4%
Austin Riley	16.7%	Kyle Schwarber	19.0%
Michael Harris II ✓	15.7%	Austin Riley	16.7%
		Michael Harris II ✓	15.4%
XGBoost · 6/6 shared 2/6 hit			
Matt Olson ✓	21.0%		
Ronald Acuña Jr.	20.6%		
Drake Baldwin	19.7%		
Kyle Schwarber	19.1%		
Austin Riley	16.4%		
Michael Harris II ✓	15.2%		

Hitter by Hitter

Hitter	Served (full season)	Flat league	KNN comps	SVR	Random forest	Core (KNN+SVR)	Forest blend	Linear regression	Logistic regression
Philadelphia Phillies · vs LHP · 46 HR / 710 PA	18.6%	17.1%	18.4%	19.2%	18.9%	18.4%	18.4%	18.8%	19.7%
5. Austin Riley Atlanta Braves · vs RHP · 20 HR / 639 PA	17.1%	15.8%	16.7%	16.7%	16.3%	16.7%	16.7%	16.5%	16.7%
6. Michael Harris II Atlanta Braves · vs RHP · 26 HR / 630 PA	15.1%	15.2%	15.1%	15.5%	14.9%	15.1%	15.1%	15.2%	15.7%
7. Bryce Harper Philadelphia Phillies · vs LHP · 30 HR / 702 PA	14.5%	14.1%	14.7%	15.3%	14.9%	14.7%	14.7%	14.9%	15.0%
8. Mike Yastrzemski Atlanta Braves · vs RHP · 12 HR / 413 PA	14.1%	13.9%	13.9%	13.7%	13.9%	13.9%	13.9%	13.6%	13.6%
9. Alec Bohm Philadelphia Phillies · vs LHP · 16 HR / 638 PA	13.4%	13.7%	13.2%	12.9%	13.2%	13.2%	13.2%	12.9%	12.9%
10. Edmundo Sosa Philadelphia Phillies · vs LHP · 9 HR / 256 PA	12.8%	12.7%	12.2%	12.4%	12.2%	12.2%	12.2%	12.4%	12.4%
11. Trea Turner Philadelphia Phillies · vs LHP · 19 HR / 709 PA	11.2%	12.8%	11.8%	12.4%	12.1%	11.8%	11.8%	12.0%	12.1%
12. Dominic Smith Atlanta Braves · vs RHP · 9 HR / 302 PA	11.1%	11.2%	10.8%	11.2%	11.0%	10.8%	10.8%	11.4%	10.7%
13. J.T. Realmuto Philadelphia Phillies · vs LHP · 11 HR / 474 PA	11.1%	11.2%	10.9%	11.0%	10.9%	10.9%	10.9%	11.0%	10.7%
14. Bryan De La Cruz Philadelphia Phillies · vs LHP · 2 HR / 92 PA	10.7%	11.9%	10.5%	10.1%	10.0%	10.5%	10.5%	9.8%	9.5%
15. Ozzie Albies Atlanta Braves · vs RHP · 22 HR / 692 PA	10.4%	10.9%	10.2%	11.1%	11.1%	10.2%	10.2%	10.5%	10.5%
16. Mauricio Dubón Atlanta Braves · vs RHP · 12 HR / 628 PA	10.0%	11.7%	10.2%	10.4%	10.4%	10.2%	10.2%	9.9%	10.4%
17. Bryson Stott Philadelphia Phillies · vs LHP · 10 HR / 617 PA	10.0%	10.0%	9.5%	8.9%	9.0%	9.5%	9.5%	9.2%	9.2%
18. Justin Crawford Philadelphia Phillies · vs LHP · 3 HR / 470 PA	5.0%	7.0%	5.3%	4.3%	4.8%	5.3%	5.3%	4.3%	5.0%

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context.

HR Daily Tracker — model comparison

Slate for 2026-10-01 · model built 2026-10-02 18:30:44 · live data 2026-10-02 18:34:09